

DEPLOYMENT TRACKER

Maaxly "Plan F" Deployment Log & Guide

This document tracks the complete, step-by-step deployment of the Maaxly project onto the GCP Free Trial, following the "Plan F" architecture.

Section 1: Initial Setup (Completed)

This section logs the initial setup of the local environment.

1.1: GCloud CLI Installation & Init

```
* **Status:** `COMPLETED`
* **Date:** 2025-11-02
* **Log:**
...
Welcome to the Google Cloud CLI!
...
You are signed in as: [maharajanabishekyt@gmail.com].
Pick cloud project to use:
[7] Create a new project
Enter a Project ID: maaxly-deploy-trial
...
Your current project has been set to: [maaxly-deploy-trial].
The Google Cloud CLI is configured and ready to use!
* Commands that require authentication will use maharajanabishekyt@gmail.com by default
* Commands will reference project maaxly-deploy-trial by default
...
```

1.2: GitHub Deploy Key Creation

```
* **Status:** `COMPLETED`
* **Date:** 2025-11-02
* **Action:** Generated a new SSH key pair (`gcp_deploy_key` and `gcp_deploy_key.pub`) locally.
* **Log:** The public key was added to the `TentacioPro/maaxly` repository.
...
GCP VM-1 App Server
SHA256:RVALDV90IWAH/dUPbEAL6a3EW8msn/ascAC78LH/M
Added on Nov 2, 2025 by @TentacioPro
Never used-Read-only
...
```

Section 2: Git & Project File Setup (To-Do)

This is your next step. You must prepare your repository *before* you create the VMs.

2.1: Define Your Git Branching Strategy

You need to create the branch that the VM will deploy from.

- * **`main`** : Your stable, primary branch.
- * **`gcp-deploy-300`** : Your **production branch**. This is the branch VM-1 will clone.
- * **`staging`** : Your **testing branch**. You will test changes here before merging them into `gcp-deploy-300`.
- * **`feature/\*`** : Your development branches (e.g., `feature/new-chat-ui`).

2.2: Create `gcp-deploy-300` Branch

On your local Windows machine, in your `maaxly` project folder:

```
```powershell
Make sure you are on your main branch and have the latest code
git checkout main
git pull

Create the new deployment branch from main
git checkout -b gcp-deploy-300

Push the new branch to GitHub
git push -u origin gcp-deploy-300
```

## 2.3: Add the **Dockerfile** (Corrected)

In the **root** of your `maaxly` project, create a new file named `Dockerfile` (no extension).

```
--- STAGE 1: Build the React Frontend ---
FROM node:20-alpine AS builder
WORKDIR /app

Copy the single root package.json and install all dependencies
COPY package.json package-lock.json* ./
RUN npm install

Copy the rest of the source code
COPY . .

Build the frontend (Vite)
RUN npm run build

--- STAGE 2: Build the Production Server ---
FROM node:20-alpine AS runner
WORKDIR /app

ENV NODE_ENV=production
```

```
Copy the single root package.json again
COPY package.json package-lock.json* ./

Install *only* the production dependencies (skips devDependencies like
vite/react)
RUN npm install --production

Copy the server source code
COPY server/ ./server/

Copy the built React app from the 'builder' stage
COPY --from=builder /app/dist ./server/dist

Expose the port your server runs on
EXPOSE 4000

The command to start your server
CMD ["node", "server/index.js"]
```

## 2.4: Commit and Push the Dockerfile

On your local Windows machine, commit the new `Dockerfile` to your `gcp-deploy-300` branch.

PowerShell

```
Add the new Dockerfile
git add Dockerfile
```

```
Commit the change to your deployment branch
git commit -m "feat: Add production Dockerfile for GCP deployment"
```

```
Push the commit
git push
```

## Section 3: Secure Private Key in GCP (Completed)

This section details uploading the private deploy key to GCP Secret Manager.

### 3.1: Enable Secret Manager API & Link Billing

- **Status:** `COMPLETED`

- **Log:** Attempted to create the secret, which prompted to enable the Secret Manager API. The API enablement failed because the new project was not linked to the free trial billing account.

```
PS C:\Users\Abishek> gcloud secrets create maaxly-github-deploy-key --replication-policy="automatic"
ERROR: (gcloud.secrets.create) FAILED_PRECONDITION: Billing account for project '718338082823' is not found...
```

- **Resolution:** Went to the [GCP Billing Console](#), found the `maaxly-deploy-trial` project, selected "Change billing," and linked it to the active free trial billing account.

## 3.2: Create Secret Container

- **Status:** `COMPLETED`

- **Log:**

```
PS E:\Other\Maaxly November GCP Deployment> gcloud secrets create maaxly-github-deploy-key --
replication-policy="automatic"
(Success after linking billing)
```

## 3.3: Add Secret Version

- **Status:** `COMPLETED`

- **Log:**

```
PS E:\Other\Maaxly November GCP Deployment> gcloud secrets versions add maaxly-github-deploy-key --
data-file="./gcp_deploy_key"
Created version [1] of the secret [maaxly-github-deploy-key].
```

## 3.4: Grant VM Access to the Secret (Error)

- **Status:** `ERROR`

- **Log:** The command to grant permissions failed because the default Compute Engine service account did not exist yet.

```
PS E:\Other\Maaxly November GCP Deployment> $SERVICE_ACCOUNT = "${PROJECT_NUMBER}-
compute@developer.gserviceaccount.com"
PS E:\Other\Maaxly November GCP Deployment> gcloud secrets add-iam-policy-binding maaxly-github-
deploy-key `
>> --member="serviceAccount:$SERVICE_ACCOUNT" `
>> --role="roles/secretmanager.secretAccessor"
ERROR: (gcloud.secrets.add-iam-policy-binding) Status code: 400. Service account 718338082823-
compute@developer.gserviceaccount.com does not exist..
```

## 3.5: Enable Compute Engine API (Fix)

- **Status:** `COMPLETED`

- **Action:** Manually enabled the Compute Engine API to force the creation of the default service account.
- **Log:** PowerShell

```
gcloud services enable compute.googleapis.com
```

### 3.6: Re-run Grant VM Access (Success)

- **Status:** COMPLETED
- **Action:** Re-ran the failed command after the API was enabled.
- **Log:**

```
PS E:\Other\Maaxly November GCP Deployment> gcloud secrets add-iam-policy-binding maaxly-github-
deploy-key --member="serviceAccount:$SERVICE_ACCOUNT" --
role="roles/secretmanager.secretAccessor"
Updated IAM policy for secret [maaxly-github-deploy-key].
bindings:
- members:
 - serviceAccount:718338082823-compute@developer.gserviceaccount.com
 role: roles/secretmanager.secretAccessor
etag: BwZCnJdghDU=
version: 1
```

## Section 4: Final VM Startup Scripts & Configs (Completed)

These are the final, corrected scripts and configurations as fixed and executed by the Gemini CLI.

### `Dockerfile` (Final)

This version corrects the build error by using the single root `package.json`.

```
```dockerfile
# --- STAGE 1: Build the React Frontend ---
FROM node:20-alpine AS builder
WORKDIR /app
COPY package.json package-lock.json* ./
RUN npm install
COPY ..
RUN npm run build

# --- STAGE 2: Build the Production Server ---
FROM node:20-alpine AS runner
WORKDIR /app
ENV NODE_ENV=production
COPY package.json package-lock.json* ./
RUN npm install --production
COPY server/ ./server/
COPY --from-builder /app/dist ./server/dist
```

```
EXPOSE 4000
CMD [ "node", "server/index.js" ]
```

server/index.js (Final Fix)

The Gemini CLI patched this file on the VM to serve the static frontend. The key fix was to add `express.static` middleware *after* all API routes.

vm-1-app docker-compose.yml (Final)

This file runs the app, the proxy, and Portainer.

YAML

```
version: '3.8'
services:
  maaxly-app:
    build:
      context: .
      dockerfile: Dockerfile
    container_name: maaxly-app
    restart: unless-stopped
    environment:
      - MONGODB_URI=mongodb://10.128.0.2:27017/maaxly-db
      - REDIS_URL=redis://10.128.0.2:6379
      - JWT_SECRET=H+cpMJchWXEYRf6sB/6XqrOkqgPEuPIIHaziSfLFmXzcKndqtW+vbzPgGiAvAY2I
      - PORT=4000
      - CORS_ORIGIN=[https://maaxly-trial.duckdns.org](https://maaxly-trial.duckdns.org)
    depends_on:
      - npm-proxy
  npm-proxy:
    image: 'jc21/nginx-proxy-manager:latest'
    container_name: npm-proxy
    restart: unless-stopped
    ports: ["80:80", "443:443", "81:81"]
    volumes: ["npm_data:/data", "npm_letsencrypt:/etc/letsencrypt"]
  portainer:
    image: portainer/portainer-ce:latest
    container_name: portainer
    restart: unless-stopped
    ports:
      - "9000:9000"
    volumes:
      - "/var/run/docker.sock:/var/run/docker.sock"
      - "portainer_data:/data"
volumes:
  npm_data:
  npm_letsencrypt:
  portainer_data:
```

vm-2-data docker-compose.yml (Final)

This file runs the database, cache, and all management UIs.

YAML

```
version: '3.8'
services:
  maaxly-mongo:
    image: mongo:latest
    container_name: maaxly-mongo
    ports: ["27017:27017"] #Exposed only to internal VPC
    volumes: ["mongo_data:/data/db"]
    restart: unless-stopped
  maaxly-redis:
    image: redis:7-alpine
    container_name: maaxly-redis
    ports: ["6379:6379"] # Exposed only to internal VPC
    volumes: ["redis_data:/data"]
    restart: unless-stopped
  portainer:
    image: portainer/portainer-ce:latest
    container_name: portainer
    restart: unless-stopped
    ports:
      - "9000:9000"
    volumes:
      - "/var/run/docker.sock:/var/run/docker.sock"
      - "portainer_data:/data"
  redis-insight:
    image: redislabs/redisinsight:latest
    container_name: redis-insight
    restart: unless-stopped
    ports:
      - "8001:8001"
    volumes:
      - "redis_insight_data:/db"
  mongo-express:
    image: mongo-express:latest
    container_name: mongo-express
    restart: unless-stopped
    ports:
      - "8081:8081"
    environment:
      - ME_CONFIG_MONGODB_SERVER=maaxly-mongo
volumes:
  mongo_data:
  redis_data:
  portainer_data:
  redis_insight_data:
```

Section 5: Provision GCP Infrastructure (Completed)

This section logs the creation of firewall rules and virtual machines.

5.1: Create Firewall Rules (Completed)

- **Status:** `COMPLETED`
- **Log:** Encountered and fixed two PowerShell formatting errors.
 1. `-allow=tcp:80,tcp:443` failed because it was not in quotes.
 2. `-target-tags=app-server,db-server` also failed for not being in quotes.
- **Resolution:** All comma-separated lists in PowerShell `gcloud` commands must be wrapped in quotes (e.g., `-allow="tcp:80,tcp:443"`, `-tags="tag1,tag2"`). The firewall rules were created successfully after this correction.

5.2: Provision VM-2 (Error)

- **Status:** `ERROR`
- **Log:** The `gcloud compute instances create vm-2-data` command failed multiple times with flag errors.
 1. `-enable-monitoring` (unrecognized)
 2. `-stack-type=monitoring` (invalid choice)
 3. `-install-ops-agent` (unrecognized)
 4. `-scopes="[https://...](https://...)"` (Invalid value). This was a repeated copy-paste error from a markdown link.
- **Resolution:**
 1. The Ops Agent installation was moved to the startup scripts in `Section 4`.
 2. The `-scopes` flag must be a plain text URL string, not a markdown link: `-scopes="<https://www.googleapis.com/auth/cloud-platform>"`.

5.3: Run Infrastructure Script (Completed)

- **Status:** `COMPLETED`
- **Log:** The fully corrected script ran successfully.

```
PS E:\\Other\\Maaxly November GCP Deployment> gcloud compute instances create vm-2-data ...
WARNING: You have selected a disk size of under [200GB]. This may result in poor I/O performance...
Created [[<https://www.googleapis.com/compute/v1/projects/maaxly-deploy-trial/zones/us-central1-a/instances/vm-2-data>](<https://www.googleapis.com/compute/v1/projects/maaxly-deploy-trial/zones/us-cent
```

```
ral1-a/instances/vm-2-data>)].
```

...

NAME	ZONE	MACHINE_TYPE	PREEMPTIBLE	INTERNAL_IP	EXTERNAL_IP	STATUS
vm-2-data	us-central1-a	e2-medium		10.128.0.2		RUNNING

```
PS E:\\Other\\Maaxly November GCP Deployment> gcloud compute addresses create maaxly-static-ip --region=us-central1
```

```
Created [[<https://www.googleapis.com/compute/v1/projects/maaxly-deploy-trial/regions/us-central1/addresses/maaxly-static-ip>](<https://www.googleapis.com/compute/v1/projects/maaxly-deploy-trial/regions/us-central1/addresses/maaxly-static-ip>)].
```

```
PS E:\\Other\\Maaxly November GCP Deployment> $env:STATIC_IP = (gcloud compute addresses describe maaxly-static-ip --region=us-central1 --format="value(address)")
```

```
PS E:\\Other\\Maaxly November GCP Deployment> Write-Host "✅ Reserved Static IP: $env:STATIC_IP. Point your DuckDNS domain to this IP." -ForegroundColor Green
```

```
✅ Reserved Static IP: 34.55.189.13. Point your DuckDNS domain to this IP.
```

```
PS E:\\Other\\Maaxly November GCP Deployment> gcloud compute instances create vm-1-app ...
```

```
WARNING: You have selected a disk size of under [200GB]. This may result in poor I/O performance...
```

```
Created [[<https://www.googleapis.com/compute/v1/projects/maaxly-deploy-trial/zones/us-central1-a/instances/vm-1-app>](<https://www.googleapis.com/compute/v1/projects/maaxly-deploy-trial/zones/us-central1-a/instances/vm-1-app>)].
```

...

NAME	ZONE	MACHINE_TYPE	PREEMPTIBLE	INTERNAL_IP	EXTERNAL_IP	STATUS
vm-1-app	us-central1-a	e2-medium		10.128.0.3	34.55.189.13	RUNNING

```
PS E:\\Other\\Maaxly November GCP Deployment> Write-Host "🎉 Deployment Complete" -ForegroundColor Green
```

oyment script finished. VMs are provisioning." -ForegroundColor Cyan
🎉 Deployment script finished. VMs are provisioning.

5.4: Manual VM Provisioning (Completed)

- **Status:** `COMPLETED`
- **Issue:** Section 5 was run before Section 4, meaning the VMs were created "plain". Manual copy-paste of scripts into the VM terminal resulted in corrupted files with markdown links.
- **Resolution (Completed):** The Gemini CLI in Cloud Shell was used to:
 1. Create the clean scripts (from `Section 4`) in the Cloud Shell environment.
 2. Troubleshoot and add `sudo` prefixes to all commands needing root access.
 3. Troubleshoot and fix the Docker GPG key installation method.
 4. Troubleshoot and temporarily add an external IP to `vm-2-data` to allow package downloads.
 5. Upload (`scp`) and execute (`ssh`) the corrected `startup-vm-2.sh` on `vm-2-data`.
 6. Remove the temporary external IP from `vm-2-data`.
 7. Upload (`scp`) and execute (`ssh`) the corrected `startup-vm-1.sh` on `vm-1-app`.
- **Log:**

Summary of Actions

7. Script Execution: ...

* Upload and execute startup-vm-2.sh on vm-2-data, which completed successfully.

* Remove the temporary external IP address from vm-2-data...

* Upload and execute startup-vm-1.sh on vm-1-app, which also completed successfully.

Section 6: Final Manual Configuration (Completed)

This section logs the final configuration of the running application.

6.1: Dockerfile Build (Error)

- **Log:** `sudo docker compose build` failed with `"/server/package.json": not found`.
- **Resolution:** The `Dockerfile` was incorrect. It was edited on `vm-1-app` to use the single root `package.json` file.

6.2: Cloud Shell Tunnel (Error)

- **Log:** `gcloud compute ssh ...` failed with `bind [::1]:...: Cannot assign requested address`.
- **Resolution:** The Gemini CLI fixed this by forcing the tunnel to use the IPv4 loopback: `gcloud compute ssh vm-1-app --zone=us-central1-a -- -L 127.0.0.1:9090:localhost:81`

6.3: Web Preview (Error)

- **Log:** Web Preview failed with `400 Bad Request` and `CORS` errors.
- **Resolution:** The Gemini CLI instructed to bypass Web Preview and run the `gcloud` command on the local Windows machine.

6.4: Application Logic (Error)

- **Log:** The live site showed `{"message": "Server is running"}` instead of the React app.
- **Resolution:** The Gemini CLI modified the `server/index.js` file inside the `maaxly-app` container to add `express.static` middleware, then restarted the container.

6.5: Final Status

- **Status:** `COMPLETED`
- **Log:**
`This final change allows your backend server to serve both the API and the React frontend, which should resolve all the issues you were facing.`
- **Next Step:** Verify the application is live at `https://maaxly-trial.duckdns.org`.

Section 7: Adding Management UIs (In Progress)

This section logs the addition of Portainer, RedisInsight, and Mongo-Express.

7.1: Update Docker Compose Files (Completed)

- **Status:** `COMPLETED`
- **Action:** The Gemini CLI updated the `docker-compose.yml` files on both VMs and committed the changes to the `gcp-deploy-300` branch (in a new `gcp/` directory).

7.2: `vm-2-data` Pull (Error)

- **Status:** `ERROR`

- **Log:** `docker compose up -d` on `vm-2-data` failed.
`✖ portainer Error Get "[https://registry-1.docker.io/v2/](https://registry-1.docker.io/v2/)": context deadline exceeded`
- **Issue:** `vm-2-data` had no internet access to pull the new images.
- **Resolution:** The Gemini CLI temporarily added an external IP to `vm-2-data`.
`gcloud compute instances add-access-config vm-2-data --zone=us-central1-a`

7.3: Start New Services (To-Do)

- **Status:** `PENDING`
- **Action:** The external IP is active on `vm-2-data`. You must now run `docker compose up` on both VMs to start the new services.
- **1. Run on `vm-2-data` (To-Do):**
 - SSH into `vm-2-data` from Cloud Shell:Bash
`gcloud compute ssh vm-2-data --zone=us-central1-a`
 - Run the compose command:Bash
`cd /opt/maaxly`
`sudo docker compose up -d`
- **2. Run on `vm-1-app` (To-Do):**
 - SSH into `vm-1-app` from Cloud Shell:Bash
`gcloud compute ssh vm-1-app --zone=us-central1-a`
 - Run the compose command:Bash
`cd /opt/maaxly`
`sudo docker compose up -d`
- **3. Secure `vm-2-data` (CRITICAL):**
 - After the pull is complete, **you must remove the temporary IP.**
 - In Cloud Shell, run:Bash
`gcloud compute instances delete-access-config vm-2-data --zone=us-central1-a`

Section 8: Activate Disaster Recovery (To-Do)

This is the **final remaining step** of the deployment. Your automated backup scripts are installed but will fail until their destination exists.

1. Create GCS Bucket (To-Do):

- In the GCP Console, go to "Cloud Storage" > "Buckets" and create a new bucket.
- Use the *exact* name you put in the backup scripts: `maaxly-backups-bucket`
- Choose "Standard" storage, and select a US-based region (e.g., `us-central1`) to align with the "Always Free" tier.
- Click "**Create**".

2. Verify:

- The cron job on `vm-2-data` will automatically run at 2 AM and upload the database dumps to this bucket. Your deployment is now complete and resilient.